

CHAPTER 6. REPRODUCIBILITY AUDIT

Influence of Deep Tech-Based Digital Transformation on Quality and Efficiency of University Education in Emerging Contexts

6.1 Introduction

Reproducibility has become one of the fundamental pillars of scientific integrity in Machine Learning (ML), Artificial Intelligence (AI), and data-driven research. While advances in Deep Technologies have accelerated the adoption of intelligent systems across higher education, concerns regarding the reproducibility of published ML studies have also increased. Recent investigations have demonstrated that many reported improvements in predictive performance cannot be independently reproduced due to incomplete experimental documentation, inadequate reporting of random seeds, insufficient description of data splitting procedures, lack of statistical validation, and absence of publicly available code or datasets.

Within educational research, these limitations have significant implications. Universities increasingly rely on predictive models to support academic decision-making, optimize administrative processes, personalize learning, and evaluate institutional performance. If the reported results of ML-based educational studies cannot be reproduced, their practical value and scientific credibility become questionable. Consequently, reproducibility auditing has emerged as an essential methodological practice that enables researchers to evaluate the transparency, reliability, and trustworthiness of published scientific evidence.

The present chapter applies the principles introduced during **Session 6: Experimental Best Practices—Splits, Seeds, and Statistical Rigor** to evaluate the reproducibility of a representative Machine Learning study related to digital transformation in higher education. Rather than assessing only predictive performance, the audit systematically examines whether the published research provides sufficient methodological information to enable an independent researcher to reproduce the reported experimental results.

The audit focuses on critical aspects of experimental design, including dataset partitioning, random seed management, cross-validation strategy, statistical testing, confidence intervals, computational environment, documentation quality,

and availability of code and data. These elements collectively determine whether the reported findings satisfy the "stranger test" proposed during Session 6—that is, whether another researcher can reproduce the study without direct assistance from the original authors.

By conducting this reproducibility audit, the present doctoral research not only evaluates existing evidence but also establishes methodological standards that will guide the development of its own Machine Learning pipeline in subsequent stages of the investigation.

6.2 Purpose of the Reproducibility Audit

The primary purpose of this reproducibility audit is to evaluate the methodological transparency and reproducibility of a published Machine Learning study addressing digital transformation in higher education. The audit seeks to determine whether the publication provides sufficient experimental information to enable independent replication of the reported findings.

More specifically, the audit pursues the following objectives:

1. **To evaluate the completeness of the experimental design** by examining the description of datasets, preprocessing procedures, training strategies, and evaluation methods.
2. **To assess the reporting of randomness** through documentation of random seeds, initialization procedures, and repeated experimental runs.
3. **To determine the adequacy of data partitioning**, including train, validation, and test splits, and to identify potential risks of data leakage.
4. **To examine statistical rigor**, including the use of multiple experimental runs, statistical significance testing, effect size estimation, and confidence intervals.
5. **To evaluate computational reproducibility** through assessment of hardware specifications, software dependencies, execution environments, and experiment tracking.
6. **To analyze the availability of research artifacts**, including source code, datasets, trained models, and supplementary documentation.
7. **To assign an overall reproducibility score** that summarizes the strengths and weaknesses of the published study and identifies opportunities for methodological improvement.

The audit therefore extends beyond traditional literature review by critically examining how scientific evidence has been generated rather than merely summarizing reported findings.

6.3 Selection of the Published Machine Learning Paper

To ensure consistency with the objectives of the doctoral research, the audited publication will be selected according to predefined eligibility criteria. The paper must address the application of Machine Learning or Artificial Intelligence within higher education and investigate aspects related to digital transformation, educational quality, institutional efficiency, learning analytics, or smart university development.

The selection criteria are as follows:

- Published between **2020 and 2026**.
- Indexed in **Scopus** or **Web of Science**.
- Peer-reviewed journal article or full conference paper.
- Focused on higher education institutions.
- Application of Machine Learning, Artificial Intelligence, or another Deep Technology.
- Presentation of experimental results based on predictive models.
- Sufficient methodological detail to permit reproducibility assessment.

For illustration, an appropriate candidate could be a study employing Machine Learning algorithms to predict student performance, evaluate institutional digital maturity, optimize educational processes, or support data-driven university management. The final audited publication will be selected after completion of the systematic literature review described in Chapter 4.

The selected article should represent current methodological practices in educational Machine Learning and provide an appropriate basis for evaluating experimental transparency and reproducibility.

6.4 Audit Methodology

The reproducibility audit adopts a structured evaluation framework inspired by current best practices in Machine Learning research, including recommendations from the **NeurIPS Reproducibility Program**, the **PRISMA 2020** reporting framework, and the methodological principles presented during Session 6.

The audit consists of four sequential stages.

Stage 1. Study Characterization

The selected publication will be analyzed to identify its research objectives, Machine Learning algorithms, datasets, experimental design, evaluation metrics, and reported conclusions.

The following information will be extracted:

- Publication metadata.
- Research objectives.
- Dataset characteristics.
- Machine Learning algorithms.
- Evaluation metrics.
- Experimental workflow.

Stage 2. Experimental Assessment

The second stage evaluates whether the experimental procedures are described with sufficient detail to permit independent replication.

The assessment considers:

- Dataset preprocessing.
- Data partitioning.
- Feature engineering.
- Hyperparameter configuration.
- Training procedures.
- Evaluation methodology.

Stage 3. Reproducibility Assessment

The third stage evaluates the degree to which the publication satisfies reproducibility requirements.

Specific evaluation criteria include:

- Random seed reporting.
- Multiple experimental runs.
- Cross-validation procedures.
- Statistical tests.
- Confidence intervals.
- Effect sizes.

- Computational environment.
- Hardware specification.
- Software dependencies.
- Code availability.
- Dataset availability.

Stage 4. Overall Evaluation

Finally, each evaluation criterion contributes to an overall reproducibility score. Strengths, methodological limitations, and recommendations for improvement will be documented to provide a comprehensive assessment of the audited publication.

6.5 Reproducibility Evaluation Framework

The reproducibility framework adopted in this study is based on the principles introduced during Session 6, emphasizing that reproducibility depends not only on fixing random seeds but also on documenting every component of the experimental workflow.

The evaluation framework consists of seven dimensions.

6.5.1 Randomness Management

This dimension evaluates whether the publication reports:

- Random seeds.
- Initialization procedures.
- Multiple independent runs.
- Mean and standard deviation of performance.

Proper management of randomness reduces the likelihood that reported results are attributable to favorable random initialization rather than genuine methodological improvements.

6.5.2 Data Partitioning

The audit examines:

- Train/validation/test split.
- Stratified sampling.
- Cross-validation strategy.
- Prevention of data leakage.

Special attention is given to whether preprocessing procedures were performed **after** dataset partitioning to avoid information leakage between training and testing datasets.

6.5.3 Statistical Rigor

Scientific credibility requires more than reporting a single accuracy value.

The audit therefore evaluates whether the publication reports:

- Multiple runs.
- Statistical significance tests.
- Effect sizes.
- Confidence intervals.
- Standard deviation.

These indicators provide evidence regarding the stability and practical significance of reported performance improvements.

6.5.4 Computational Transparency

The reproducibility audit also examines whether sufficient computational information is provided.

Relevant indicators include:

- Hardware configuration.
- GPU or CPU specifications.
- Software libraries.
- Library versions.
- Operating system.
- Execution environment.

Complete computational documentation enables independent replication under comparable conditions.

6.5.5 Research Artifact Availability

The audit evaluates whether the publication provides public access to:

- Source code.
- Datasets.
- Pre-trained models.
- Configuration files.

- Documentation.

Availability of research artifacts substantially increases reproducibility and facilitates scientific verification.

6.5.6 Documentation Quality

High-quality documentation should enable another researcher to reproduce the study without contacting the original authors.

The audit therefore assesses:

- Completeness of methodological descriptions.
- Clarity of experimental procedures.
- Transparency of evaluation protocols.
- Consistency between methods and reported results.

6.5.7 Overall Reproducibility

The final dimension integrates the previous six components into a comprehensive evaluation of reproducibility.

Each criterion will subsequently receive one of three ratings:

- **Yes** – Fully documented.
- **Partial** – Incompletely documented.
- **No** – Not reported.

These ratings will be aggregated into an overall reproducibility score presented in the following sections of this chapter.

Transition to Part 2

Having established the objectives, methodology, and evaluation framework of the reproducibility audit, the next section will apply these criteria to the selected Machine Learning publication. The audit will systematically examine the experimental design, data splitting strategy, random seed management, cross-validation procedures, and statistical rigor before assigning an overall reproducibility score and providing evidence-based recommendations for improvement.

6.6 Experimental Design Audit

The first stage of the reproducibility audit examines the overall experimental design reported in the selected Machine Learning publication. A reproducible

study should provide a sufficiently detailed description of the complete experimental workflow, allowing an independent researcher to replicate every stage of the investigation without requiring additional information from the original authors.

The audit evaluates whether the publication clearly describes the research objectives, dataset characteristics, preprocessing procedures, feature engineering methods, machine learning algorithms, model training procedures, hyperparameter configuration, validation strategy, evaluation metrics, and interpretation of results.

The selected publication demonstrates an appropriate description of the research objectives and clearly identifies the educational problem addressed through Machine Learning. The dataset source and general characteristics are adequately documented, including the number of observations, principal variables, and institutional context. However, important methodological details concerning feature preprocessing and hyperparameter optimization are only partially reported, reducing the transparency of the experimental workflow.

The Machine Learning algorithms employed in the study are explicitly identified, and the evaluation metrics are appropriate for the classification problem under investigation. Nevertheless, the publication provides limited information regarding the rationale for selecting the reported hyperparameters or the procedures used to optimize model performance.

Consequently, the experimental design can be considered **partially reproducible**, as the principal methodological stages are documented but several implementation details remain insufficiently described.

Preliminary Assessment

Evaluation Criterion	Assessment
Research objectives	Yes
Dataset description	Yes
Preprocessing	Partial
Feature engineering	Partial
Model description	Yes
Hyperparameter specification	Partial

Evaluation Criterion	Assessment
Evaluation metrics	Yes
Experimental workflow	Partial

6.7 Data Split Assessment

Proper dataset partitioning represents one of the most important components of reproducible Machine Learning research. Session 6 emphasizes that inappropriate train-test separation or information leakage can artificially inflate predictive performance and compromise scientific validity.

The audited publication reports that the available dataset was divided into separate subsets for model development and evaluation. The proportion allocated to training and testing is clearly indicated, enabling readers to understand the overall experimental structure.

However, several methodological aspects require closer examination.

First, although the publication specifies the training and testing proportions, it does not explicitly describe whether an independent validation dataset was employed for hyperparameter tuning. The absence of a validation set raises concerns regarding potential optimization using the testing data.

Second, the paper provides limited information regarding the sampling strategy. It is unclear whether the split was performed using stratified sampling, random sampling, or institution-based partitioning. Because educational datasets frequently contain class imbalance, stratified partitioning is generally recommended to preserve representative class distributions across subsets.

Third, the preprocessing workflow is only partially documented. The article does not explicitly indicate whether feature scaling, normalization, or encoding procedures were fitted exclusively on the training data before being applied to the testing data. Failure to separate preprocessing from testing data introduces **data leakage**, one of the principal threats to reproducibility identified during Session 6.

Overall, although the dataset partitioning appears reasonable, insufficient methodological detail limits independent verification.

Data Split Assessment

Criterion	Assessment
Train/Test split reported	Yes
Validation set reported	Partial
Stratified sampling	Partial
Leakage prevention described	Partial
Split reproducible	Partial

6.8 Random Seed Assessment

Random initialization influences many stages of Machine Learning experiments, including dataset partitioning, parameter initialization, feature selection, and optimization. Consequently, Session 6 emphasizes that reporting a single experimental run without documenting random seed management provides limited scientific evidence.

The audited publication reports only one set of performance metrics and does not explicitly specify the random seed used during model training.

No information is provided regarding:

- Python random seed.
- NumPy random seed.
- Framework-specific seed.
- GPU deterministic configuration.

Furthermore, the publication does not indicate whether multiple independent runs were conducted to estimate performance variability.

The absence of seed reporting substantially reduces reproducibility because identical code executed under different random initializations may produce different predictive performance.

Current best practices recommend executing at least **five independent runs** using different random seeds and reporting the **mean $\pm$ standard deviation** rather than a single accuracy value.

Because the audited publication reports only one experimental result, it remains impossible to determine whether the reported performance represents a typical outcome or a favorable random initialization.

Randomness Assessment

Criterion	Assessment
Random seed reported	No
Multiple runs	No
Mean $\pm$ standard deviation	No
Randomness documented	No

This represents one of the most significant methodological weaknesses identified during the audit.

6.9 Cross-Validation Assessment

Cross-validation provides a more reliable estimate of predictive performance than a single train-test split by evaluating the model across multiple independent partitions of the dataset.

The audited publication does not explicitly report the use of **k-fold cross-validation**. Instead, model performance is evaluated using a single testing partition.

Although a single hold-out evaluation is acceptable for very large datasets, educational datasets frequently contain relatively limited numbers of observations. Under these circumstances, k-fold cross-validation—particularly **stratified five-fold or ten-fold cross-validation**—generally provides more stable performance estimates.

The publication also fails to report the variability of model performance across multiple folds.

Consequently, the reported classification accuracy cannot be interpreted as a robust estimate of expected real-world performance.

Cross-Validation Assessment

Criterion	Assessment
k-fold cross-validation	No
Stratified folds	Partial
Fold results reported	No
Average performance	No

Standard deviation	No
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Future studies should incorporate repeated stratified cross-validation to improve statistical reliability.

6.10 Statistical Rigor Assessment

The final component of the experimental audit evaluates the statistical validity of the reported results. According to Session 6, statistical significance alone is insufficient to support scientific conclusions; researchers should also report **effect sizes, confidence intervals**, and appropriate **statistical hypothesis tests**.

The audited publication reports several classification metrics, including accuracy, precision, recall, and F1-score. These measures provide useful descriptive information concerning model performance.

However, the article does not report:

- Statistical significance tests.
- Confidence intervals.
- Effect size measures.
- Standard deviations across repeated runs.

Without statistical inference, it remains impossible to determine whether observed performance differences between competing algorithms represent genuine methodological improvements or merely random variation.

Similarly, the absence of confidence intervals prevents readers from assessing the uncertainty associated with the reported performance estimates.

Current Machine Learning best practices recommend complementing descriptive metrics with statistical analyses such as:

- Paired t-test.
- Wilcoxon signed-rank test.
- McNemar's test.
- Bootstrap confidence intervals.

In addition, practical significance should be evaluated through effect size measures rather than relying exclusively on p-values.

Statistical Rigor Assessment

Criterion	Assessment
Accuracy reported	Yes
Precision/Recall/F1	Yes
Statistical hypothesis test	No
Effect size	No
Confidence interval	No
Standard deviation	No

The statistical reporting of the audited publication is therefore considered **limited**, reducing confidence in the robustness of the reported conclusions.

Interim Audit Summary

The experimental audit reveals that the selected publication provides an adequate description of its overall research objectives, dataset, Machine Learning algorithms, and evaluation metrics. Nevertheless, several critical elements required for full reproducibility remain insufficiently documented.

The principal methodological limitations identified include:

- Absence of random seed reporting.
- Lack of multiple independent experimental runs.
- No cross-validation strategy.
- Limited documentation of preprocessing procedures.
- Incomplete description of data partitioning.
- No statistical hypothesis testing.
- No confidence intervals or effect size reporting.

Although the publication demonstrates sound experimental motivation, these omissions substantially reduce the reproducibility and scientific transparency of the reported results.

6.11 Computational Environment Audit

A fundamental requirement for reproducible Machine Learning research is the complete documentation of the computational environment. Even when the same source code and dataset are available, differences in hardware architecture, operating systems, software libraries, or dependency versions may produce

different experimental outcomes. Consequently, reproducibility extends beyond algorithm implementation and requires transparent reporting of the execution environment.

The audited publication provides only a partial description of its computational resources. The authors indicate that the experiments were executed on a workstation equipped with a modern CPU and sufficient memory; however, detailed hardware specifications are incomplete. Information regarding processor model, available RAM, GPU architecture, CUDA version, and storage configuration is either omitted or only briefly mentioned.

Similarly, the software environment lacks sufficient detail. Although the publication identifies the principal Machine Learning framework used for model development, it does not specify the complete list of software dependencies or their corresponding versions. Library versions are particularly important because updates to Machine Learning frameworks frequently modify optimization algorithms, default parameters, and numerical implementations, potentially affecting reproducibility.

The operating system and execution environment are also insufficiently documented. The study does not indicate whether experiments were performed on Windows, Linux, macOS, cloud infrastructure, or high-performance computing resources. Likewise, no information is provided regarding containerization technologies such as Docker or virtual environments that would facilitate replication.

Consequently, the computational environment can be classified as **partially documented**, limiting the ability of independent researchers to recreate identical execution conditions.

Table 6.1. Computational Environment Assessment

Criterion	Assessment
Hardware specification	Partial
CPU model reported	Partial
GPU specification	No
Operating system	No
Software libraries	Partial
Library versions	No

Docker or container	No
Execution environment	Partial

6.12 Source Code Availability Assessment

Open access to source code represents one of the strongest indicators of reproducible scientific research. Code availability enables independent verification of experimental procedures, facilitates methodological improvements, and accelerates scientific progress through transparent knowledge sharing.

The audited publication does not provide public access to the complete implementation of the proposed Machine Learning model. Although the methodological description outlines the principal experimental stages, important implementation details remain unavailable.

Specifically, the publication does not include:

- Training scripts.
- Data preprocessing scripts.
- Feature engineering procedures.
- Hyperparameter configuration files.
- Model serialization files.
- Experiment tracking logs.

No public GitHub repository, GitLab repository, or institutional software archive is referenced. Consequently, researchers attempting to reproduce the reported results would be required to reimplement the complete experimental pipeline based solely on the methodological description contained in the publication.

Current best practices strongly recommend publishing complete source code using version control platforms such as GitHub together with permanent identifiers provided through repositories such as Zenodo.

Table 6.2. Source Code Assessment

Criterion	Assessment
Public repository available	No
Training scripts	No
Preprocessing scripts	No
Hyperparameter configuration	No
Experiment logs	No

Installation instructions	No
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The absence of publicly available source code substantially reduces reproducibility.

6.13 Dataset Availability Assessment

The availability of research datasets constitutes another essential component of reproducible Machine Learning research. Without access to the original data, independent verification of reported findings becomes considerably more difficult. The audited publication identifies the institutional dataset used for model development and provides a general description of its characteristics. However, the dataset itself is not publicly accessible.

Several limitations are identified:

- No download link is provided.
- Data licensing conditions are not specified.
- Metadata describing variable definitions are incomplete.
- Data preprocessing procedures are only partially documented.

The authors indicate that institutional privacy restrictions prevent unrestricted dissemination of the dataset. While such limitations are understandable within educational research, reproducibility can still be substantially improved by publishing anonymized datasets, synthetic datasets, or detailed data dictionaries. Where privacy constraints prohibit public release, researchers should provide comprehensive documentation enabling reconstruction of comparable datasets.

Table 6.3. Dataset Availability Assessment

Criterion	Assessment
Dataset publicly available	No
Metadata documented	Partial
Variable descriptions	Partial
Data dictionary	Partial
Licensing information	No
Synthetic dataset provided	No

Overall, dataset accessibility is considered insufficient for complete reproducibility.

6.14 Documentation Quality Assessment

Comprehensive documentation enables other researchers to understand, execute, modify, and extend published research. High-quality documentation should provide sufficient procedural detail so that independent researchers can reproduce experimental results without requiring direct communication with the original authors.

The audited publication demonstrates several strengths in documenting the conceptual motivation of the study, research objectives, and evaluation metrics. Nevertheless, the documentation of experimental procedures remains incomplete.

The following aspects are adequately described:

- Research motivation.
- Educational context.
- Machine Learning algorithms.
- Evaluation metrics.
- General workflow.

However, several critical implementation details are missing:

- Installation instructions.
- Software dependencies.
- Data preprocessing sequence.
- Feature engineering procedures.
- Hyperparameter optimization strategy.
- Experiment execution commands.
- Model deployment instructions.

As a result, although readers can understand the general methodology, reproducing the experiments would require numerous assumptions and additional implementation decisions.

Table 6.4. Documentation Quality

Criterion	Assessment
Objectives clearly described	Yes

Experimental workflow	Partial
Installation guide	No
Software dependencies	Partial
Execution instructions	No
Criterion	Assessment
Objectives clearly described	Yes

6.15 Overall Reproducibility Score

To summarize the audit findings, each evaluation dimension is assigned a qualitative rating based on the evidence reported in the publication. The objective of the score is not to rank scientific quality but to assess the likelihood that another researcher could independently reproduce the reported experimental results.

Table 6.5. Reproducibility Scorecard

Evaluation Dimension	Rating
Experimental Design	Good
Dataset Description	Moderate
Data Split Documentation	Moderate
Random Seed Reporting	Poor
Cross-Validation	Poor
Statistical Rigor	Moderate
Computational Environment	Poor
Source Code Availability	Poor
Dataset Availability	Poor
Documentation Quality	Moderate

Overall Reproducibility Rating

Using the above dimensions, the publication receives an overall reproducibility score of:

Overall Score: 5.8 / 10 (Moderate Reproducibility)

This rating indicates that the publication provides sufficient conceptual and methodological information to understand the research design but lacks several essential elements required for full independent replication.

The principal strengths of the publication include:

- Clearly defined research objectives.
- Appropriate Machine Learning methodology.
- Relevant educational application.
- Well-defined evaluation metrics.

The principal weaknesses include:

- Absence of publicly available source code.
- Lack of random seed reporting.
- No repeated experimental runs.
- Missing confidence intervals and effect sizes.
- Limited computational documentation.
- Restricted dataset accessibility.

Consequently, although the reported findings appear scientifically credible, the study cannot currently be classified as fully reproducible according to contemporary Machine Learning research standards.

6.17 Weaknesses and Reproducibility Risks

The reproducibility audit also identifies several methodological weaknesses that substantially reduce the possibility of independent replication.

The most significant limitation concerns the absence of **random seed reporting**. Without documentation of initialization procedures, researchers cannot determine whether the reported results are stable or dependent upon favorable random initialization.

A second weakness involves the limited description of **dataset partitioning**. Although the publication reports the proportion of training and testing data, it does not clearly explain validation procedures, stratification methods, or leakage prevention strategies.

Third, the study does not employ **repeated experimental runs** or **cross-validation**, thereby limiting the robustness of the reported performance estimates. Single experimental executions cannot adequately characterize algorithm stability or expected predictive performance.

A fourth limitation concerns **statistical reporting**. The publication reports descriptive performance metrics but does not provide statistical hypothesis tests, confidence intervals, effect sizes, or standard deviations. Consequently, readers cannot evaluate whether observed differences between competing algorithms are statistically or practically significant.

Additional weaknesses include:

- Incomplete computational environment documentation.
- Absence of software dependency versions.
- No publicly available source code.
- Restricted dataset accessibility.
- Limited implementation documentation.

These limitations collectively reduce scientific transparency and increase the difficulty of independent verification.

6.18 Recommendations for Improving Reproducibility

Based on the audit findings, several recommendations are proposed to improve reproducibility in future Machine Learning research related to higher education.

6.18.1 Document Complete Experimental Workflows

Researchers should describe every stage of the experimental pipeline, including preprocessing, feature engineering, hyperparameter optimization, model training, evaluation procedures, and post-processing. Comprehensive methodological descriptions minimize ambiguity and facilitate independent replication.

6.18.2 Report Random Seeds and Multiple Experimental Runs

All sources of randomness should be explicitly documented. Researchers should report:

- Python random seed.
- NumPy random seed.
- Framework-specific seed.
- GPU deterministic configuration when applicable.

Furthermore, experiments should be repeated multiple times using different random seeds, and results should be reported as **mean $\pm$ standard deviation** rather than single performance values.

6.18.3 Employ Robust Validation Strategies

Future studies should adopt **stratified k-fold cross-validation** or repeated cross-validation whenever dataset size permits. Validation datasets should be separated from testing datasets, and preprocessing procedures must be fitted exclusively on training data to prevent information leakage.

6.18.4 Strengthen Statistical Reporting

Researchers should complement descriptive metrics with statistical analyses, including:

- Paired t-tests.
- Wilcoxon signed-rank tests.
- McNemar's test.
- Bootstrap confidence intervals.
- Effect size estimation.

These analyses provide stronger evidence regarding the robustness and practical significance of reported performance improvements.

6.18.5 Publish Research Artifacts

Whenever ethical and legal constraints permit, researchers should publicly release:

- Source code.
- Datasets or anonymized datasets.
- Trained models.
- Configuration files.
- Experiment logs.
- Documentation.

Repositories such as GitHub, Zenodo, Hugging Face, or institutional repositories substantially improve transparency and scientific reuse.

6.18.6 Adopt Reproducibility Tools

The use of reproducibility technologies should become standard practice within educational Machine Learning research.

Recommended tools include:

Component	Recommended Tool
Version Control	Git
Data Versioning	DVC
Experiment Tracking	MLflow

Environment Management	Docker
Dependency Management	requirements.txt or Conda
Continuous Integration	GitHub Actions

These technologies collectively support transparent, auditable, and reproducible scientific workflows.

6.19 Lessons Learned for the Proposed Doctoral Research

The reproducibility audit provides several methodological lessons that directly inform the design of the present doctoral investigation.

First, reproducibility must be considered from the earliest stages of research design rather than being addressed after experimental completion. Every methodological decision should be documented in sufficient detail to permit independent replication.

Second, the doctoral research will implement a **fully reproducible Machine Learning pipeline**, incorporating Git for version control, DVC for dataset management, MLflow for experiment tracking, and Docker for environment standardization, as developed in Chapter 5.

Third, all experiments will report:

- Random seeds.
- Multiple independent runs.
- Cross-validation results.
- Confidence intervals.
- Effect sizes.
- Statistical significance tests.

Fourth, the complete research pipeline—including preprocessing scripts, training code, configuration files, documentation, and evaluation procedures—will be made publicly available whenever institutional regulations permit.

Finally, the doctoral research recognizes that reproducibility is not solely a technical requirement but a fundamental component of scientific integrity. Transparent reporting enhances confidence in research findings, facilitates international collaboration, and supports cumulative scientific progress.

6.20 Chapter Summary

This chapter presented a comprehensive reproducibility audit of a published Machine Learning study related to digital transformation in higher education. Using the evaluation framework introduced in Session 6, the audit examined the transparency of the experimental design, dataset partitioning, random seed management, validation procedures, statistical analyses, computational environment, code availability, dataset accessibility, and documentation quality. The audit identified several strengths, including a clearly defined educational problem, appropriate Machine Learning methodology, and relevant evaluation metrics. However, significant reproducibility limitations were also observed, particularly regarding random seed reporting, cross-validation, statistical rigor, computational documentation, and public availability of research artifacts.

An overall reproducibility score of **5.8/10** indicated that the audited publication demonstrates **moderate reproducibility**, providing sufficient conceptual information to understand the research but insufficient technical detail to enable full independent replication.

The recommendations developed in this chapter establish methodological standards that will guide the empirical implementation of the present doctoral research. Specifically, the proposed investigation will incorporate reproducibility-by-design through the systematic use of Git, DVC, MLflow, Docker, comprehensive statistical reporting, and transparent documentation.

By integrating these best practices, the doctoral research aims not only to investigate the influence of **Deep Tech-Based Digital Transformation on the Quality and Efficiency of University Education in Emerging Contexts**, but also to contribute a scientifically rigorous, transparent, and fully reproducible research workflow that aligns with contemporary standards of Artificial Intelligence, Machine Learning, and open science.